



Secure Network Design (Core)

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Coding Dojo Cybersecurity Boot Camp
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Learning Objectives:

- Identify security design flaws from a network diagram.
- Provide security solutions for security design flaws.

For **each** of the following figures below:

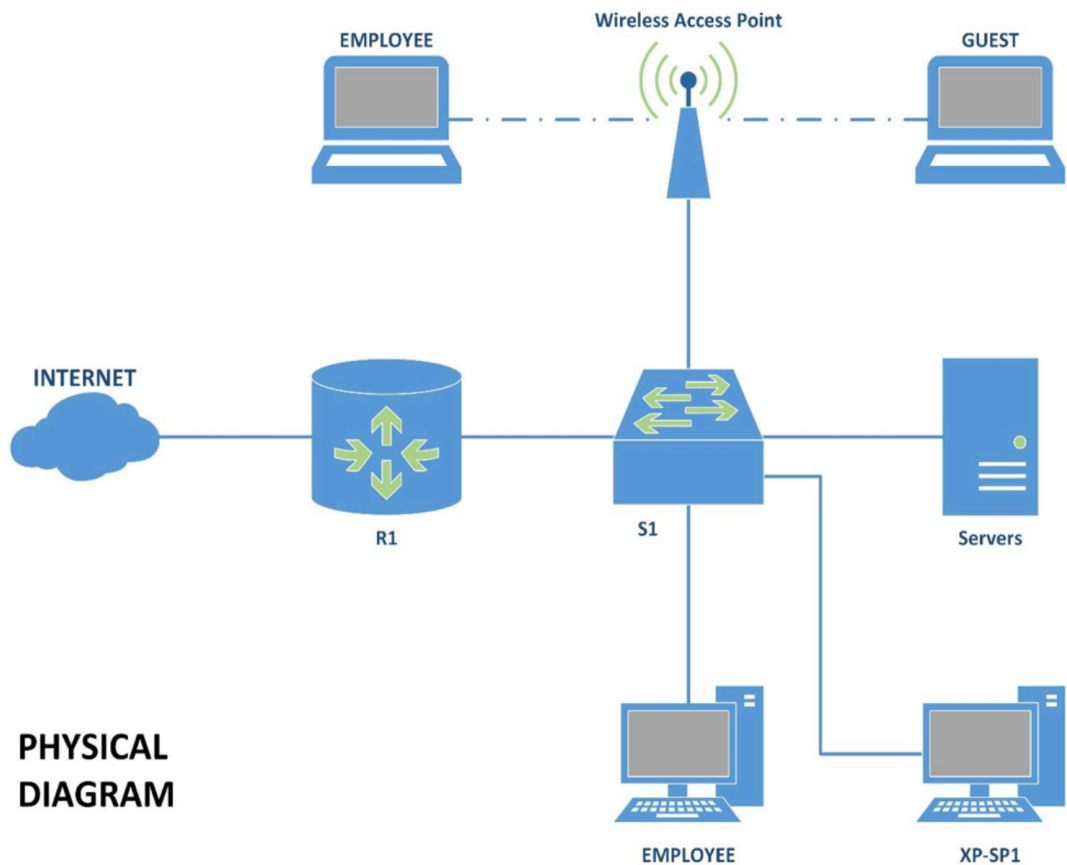
- Give at least two sentences (per figure) explaining any security concerns you may or may not have.
- Suggest security solutions that address the concerns you mentioned in #1.

Assignment Submission

Upload a single document with your responses to **each** of the network diagrams below.

Assignment Solution:

Diagram 1:



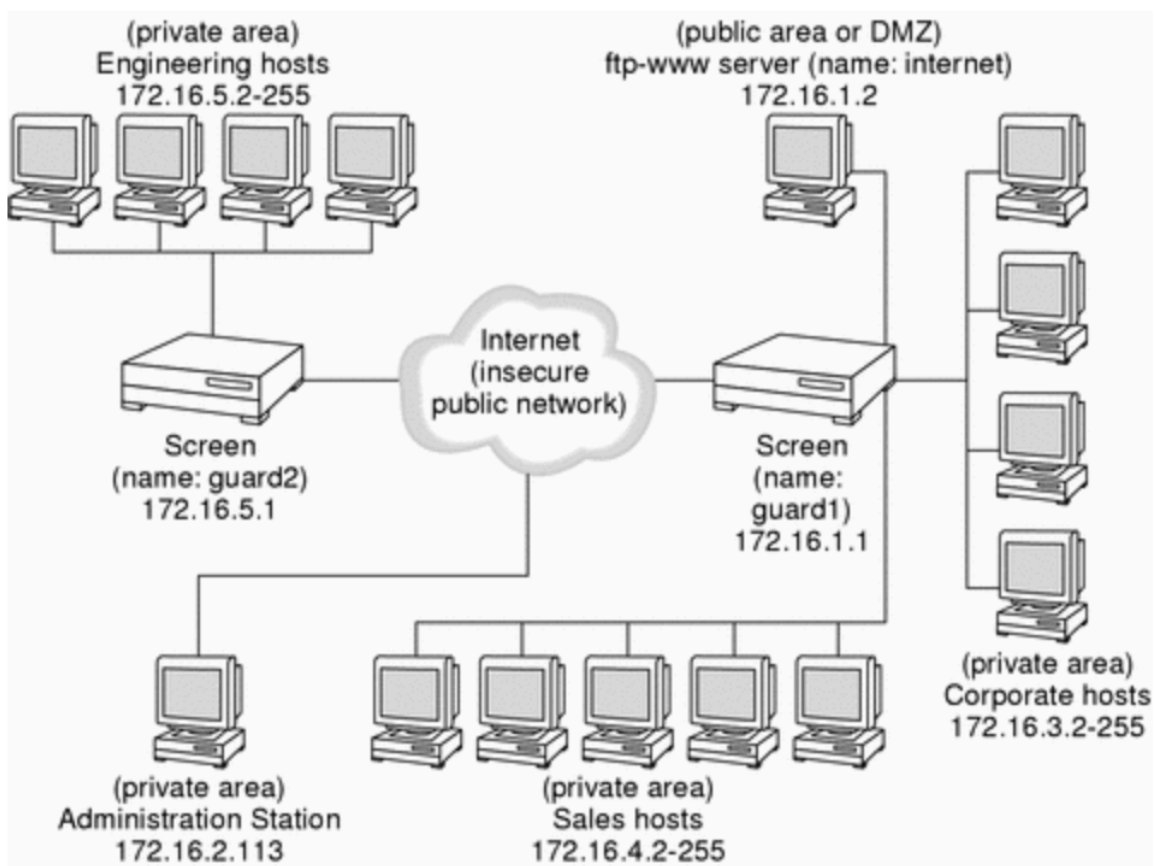
Security concerns

1. A system accessible to guest shares the same access point as an employee. This may constitute a security breach as it means data can be accessible between the 2 systems. Having both employee and guest systems connected to the same wireless access point may pose a security risk.
2. There's a system labelled XP-SP1, which probably connotes that the Operating system running on it is XP. This is an old operating system that has reached the End of Life. This opens up the network to threat actors as it may have vulnerabilities that are no longer supported or for which no patches are available to aid remediation.

Security solutions that address the concerns mentioned

1. Solutions to the security concerns raised in 1 above include:
 - a. **Segmentation and VLANs:** Guest systems should be isolated from internal resources to prevent unauthorized access.
Also, the employee system should not share access point as the same level as a guest machine; Implement VLANs to separate traffic from employee and guest systems, improving network segmentation.
 - b. **Enhanced Wireless Security:** Configure the wireless access point with strong encryption and enable features like guest isolation.
 - c. **Access Control Policies:** Define and enforce access control policies to restrict access to critical resources based on user roles and permissions.
2. Possible solutions to the security concerns raised in 2 above include:
 - a. **Discard/Upgrade XP System:** The XP system should be discarded from within the network. OR upgraded to a more recent and supported operating system.

Diagram 2:



Security concerns

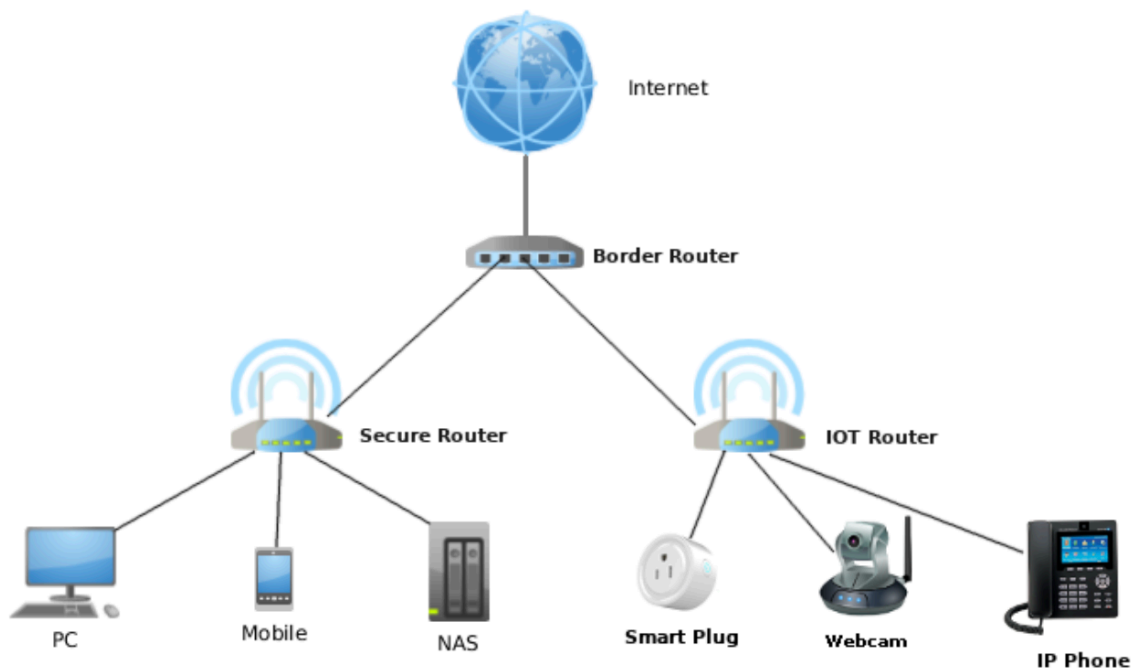
1. The administration station has direct access to the internet on an insecure public network. This raises concerns about the security of the system, as direct exposure to the internet without proper safeguards can lead to various security threats.
2. Both 'Guard1' and 'Guard2' switches, are directly connected to the internet on an insecure public network. This raises concerns about direct exposure to external threats, increasing the risk of unauthorized access, attacks, and exploitation.
3. The use of bus topology for internal networks (Engineering, Sales, Corporate hosts) can lead to a single point of failure and makes it easier for an attacker to capture and manipulate network traffic. Additionally, it seems to lack segmentation, allowing lateral movement if a compromise occurs.

4. The ftp-www server in the DMZ is a potential target for attacks. Ensuring proper configuration, restricting unnecessary services, and implementing strong access controls are essential to mitigate security risks.
5. The absence of clear network segmentation between different functional areas (Engineering, Sales, Corporate) raises concerns.

Security solutions that address the concerns mentioned above:

1. The administration station should go through a 'screen' to connect to the internet. Applicable firewall should be in place.
2. **Network Segmentation:** Implement network segmentation to isolate different functional areas (Engineering, Sales, Corporate) and prevent lateral movement. Use VLANs and firewalls to control communication between segments. Consider placing critical systems in a separate segment with stricter access controls.
3. **Implement strong firewall** rules on 'Guard1' and 'Guard2' switches, to control incoming and outgoing traffic. Allow only necessary services and applications to minimize the attack surface.
4. **Encryption for Internet Connections:**
Enable encryption for internet connections on 'Guard1' and 'Guard2' to protect data in transit. Use protocols like HTTPS and secure VPN connections.

Diagram 3:



Security concerns

1. There are no identified security concerns.

Security solutions that address the concerns mentioned

While the network design seems okay, it is imperative in order to ensure effectiveness, that the security measures undergo continuous review & management, which includes:

1. **Regular Audits & Assessments:** Conduct regular security audits and assessments to identify any potential weaknesses or vulnerabilities that may arise over time.
2. **IoT Device Security:** Keep a keen eye on the security of IoT devices. Regularly check for firmware updates and security patches provided by the manufacturers.
3. **Security Monitoring:** Implement robust security monitoring and Intrusion Detection system (IDS) to detect and respond to any anomalous or suspicious activities on the network.
4. **Secure Configurations:** Ensure that all devices, including routers and IoT devices, are configured securely. Default credentials should be changed, unnecessary services disabled, and access controls properly configured.